



α -Methylene and Glycidic Acid Analogs of IPPA as Potential Myocardial Imaging Agents

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Using organozinc cross-coupling reactions, two radiolabeled analogs of 15-(*p*-iodophenyl) pentadecanoic acid (IPPA) have been designed and synthesized as potential scintigraphic imaging agents for the heart. Both 15-(4-iodophenyl)-tridecylglycidic acid and 15-(4-iodophenyl)-2-methylene pentadecanoic acid were synthesized and radioiodinated. In tissue biodistribution studies in rats, only the α -methylene derivative of IPPA displayed a consistently higher heart to blood ratio and a substantially lower degree of thyroid accumulation than did IPPA alone. With respect to a scintigraphic imaging efficacy, the α -methylene analog of IPPA and IPPA itself showed essentially equivalent cardiac imaging profiles in rabbits, with a slight extension in imaging time for the α -methylene analog of IPPA.

Introduction

Radiodiagnostic imaging is a clinically useful technique with applications in the detection and evaluation of acute and chronic cardiovascular disorders. Several radiopharmaceuticals currently in use as cardiac imaging agents have recently been reviewed (Chilton *et al.*, 1990; Kulkarni and Corbett, 1990). Since free fatty acids (FFA) in the circulation are efficiently extracted by the myocardium and represent the principal source of energy for this essential organ, a number of investigations have examined the potential of radiolabeled fatty acids and their analogs to serve as cardiac imaging agents.

One potentially suitable imaging agent is [¹¹C]-palmitate, not only because of its efficient extraction into the myocardium, but also because the radio-nuclide does not alter the biochemical properties of the molecule (Saha *et al.*, 1992). Although [¹¹C]-palmitate has favorable imaging and biodistribution properties, it has not been widely used in nuclear medicine. The high costs of production, rapid decay of ¹¹C ($t_{1/2}$ = 20 min) and the expense of positron cameras have limited its clinical utility.

Fatty acids labeled with other gamma-emitting nuclides have also been proposed for use as myo-

cardial agents (Poe *et al.*, 1977, 1976). ¹²³I-16-iodo-9-hexadecenoic acid (Machulla *et al.*, 1978) and the saturated homolog ¹²³I-17-iodoheptadecanoic acid were among the first of a series of fatty acid analogs to show promise as imaging agents directed specifically to the heart. However, like most other compounds containing aliphatic iodide, these agents underwent rapid *in vivo* deiodination, resulting in a time-dependent increase in the background activity and a loss of tissue specificity (Van der Vusse *et al.*, 1987; Freundlieb *et al.*, 1980).

The search for a radioiodinated fatty acid that avoids the problems associated with high background levels but allows for the *in vivo* determination of myocardial metabolism lead to the design and development of ¹²³I-15-(*p*-iodophenyl) pentadecanoic acid (IPPA) (Machulla *et al.*, 1980). Because the iodine in IPPA is aromatic bound, the extent of deiodination is substantially reduced. Accordingly, background levels in images of the heart using IPPA are low. IPPA has also proven to be more useful for obtaining static myocardial scintiscans, since the scintiscan can be conveniently obtained in several projections (Kulkarni *et al.*, 1990). For these reasons, IPPA has become a widely used agent for cardiac scintigraphic imaging.

Other fatty acid analogs labeled with ¹¹C and ¹²³I have been designed to be more slowly metabolized by β -oxidation. 15-(*p*-iodophenyl)-6-tellurapentadecanoic acid (TPPA) (Goodman *et al.*, 1982), 1-(¹¹C)-3-methyl heptadecanoic acid (Livini *et al.*, 1982),

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15-(*p*-iodophenyl)-3-*R,S*-methyl pentadecanoic acid (Goodman *et al.*, 1984) and 15-(*p*-iodophenyl)-3,3-dimethyl pentadecanoic acid (DMIPPA) (Knapp *et al.*, 1986) are among such agents. Of these agents, only DMIPPA displays favorable imaging properties. Studies in dogs have shown that the peak myocardial uptake of DMIPPA is somewhat lower than IPPA. However, unlike IPPA, the peak radioactivity remains constant for a longer period of time and achieves myocardial-to-blood ratios of greater than 10:1 (Sloof *et al.*, 1993). Finally, the use of 14 (*R,S*)-[^{18}F]Fluoro-6-thia-heptadecanoic acid (FTHA) has also recently been described (De Grado *et al.*, 1991) as another potential method for trapping and concentrating a radiohalogenated fatty acid in the myocardium for cardiac imaging. This compound appears to be readily adaptable to ^{123}I and SPECT imaging.

The primary objective of this work was to design and synthesize radioiodinated analogs of long-chain fatty acids that would be efficiently extracted by the heart and then remain in the myocardium for a sufficiently long period to allow for single photon emission computerized tomography (SPECT) imag-

ing of the heart. The pharmacologic action of certain substituted glycidic acids provided a logical starting point for the design of new agents. For example, tetradecylglycidic acid (TDGA) is an analog of palmitic acid that is transported to different organs by the same mechanism as other long chain fatty acids. TDGA also serves as a substrate for acylCoA synthetase (Tutwiler *et al.*, 1981). However, glycidic acids in general, as their acylCoA derivatives, inhibit carnitine acyltransferase 1 (CAT 1), the enzyme responsible for conversion of fatty acylCoA to fatty acylcarnitine (Fig. 1). Studies have shown TDGA to be an irreversible, active site-directed inhibitor of CAT 1 (Kiorpes *et al.*, 1984). It was anticipated that iodophenylalkylglycidic acid analogs of IPPA (e.g. compound 11, Scheme 1) might be similarly taken up by the myocardium and experience a delay in their clearance from the heart through covalent association with CAT 1. At the same time, the comparatively stable iodophenyl moiety of these analogs would provide the imaging signal without significant loss of iodine to non-target tissues. In addition, an α -methylene analog (e.g. compound 8, Scheme 1), necessary as a precursor for

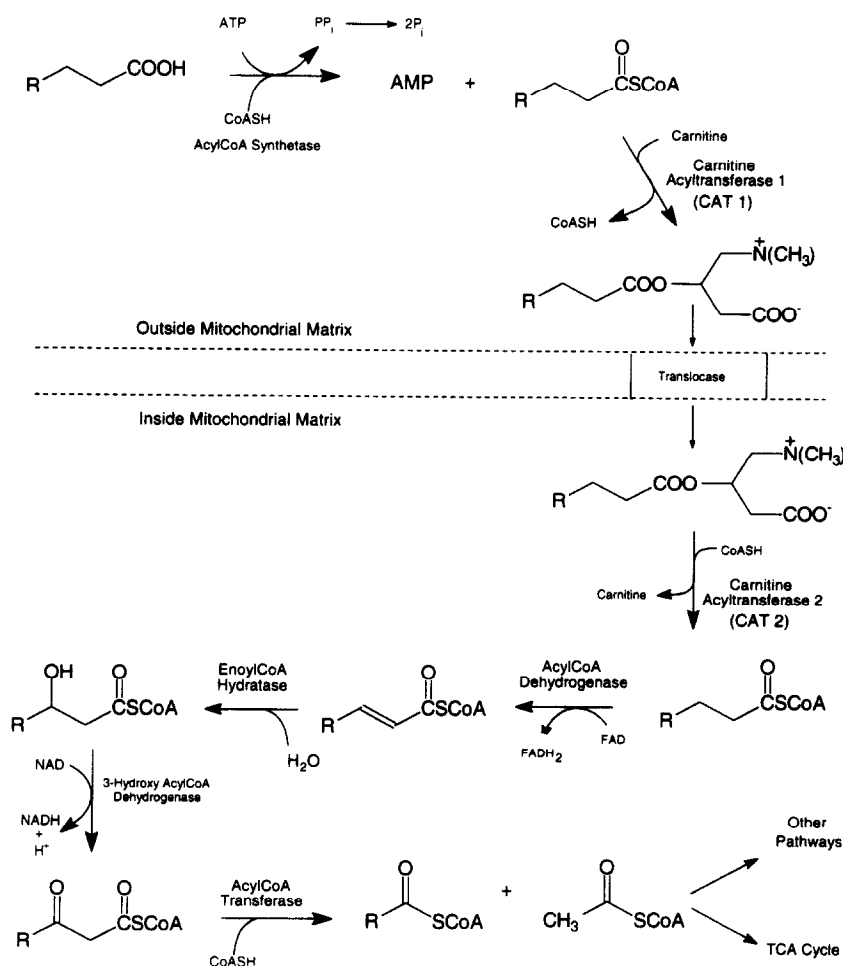
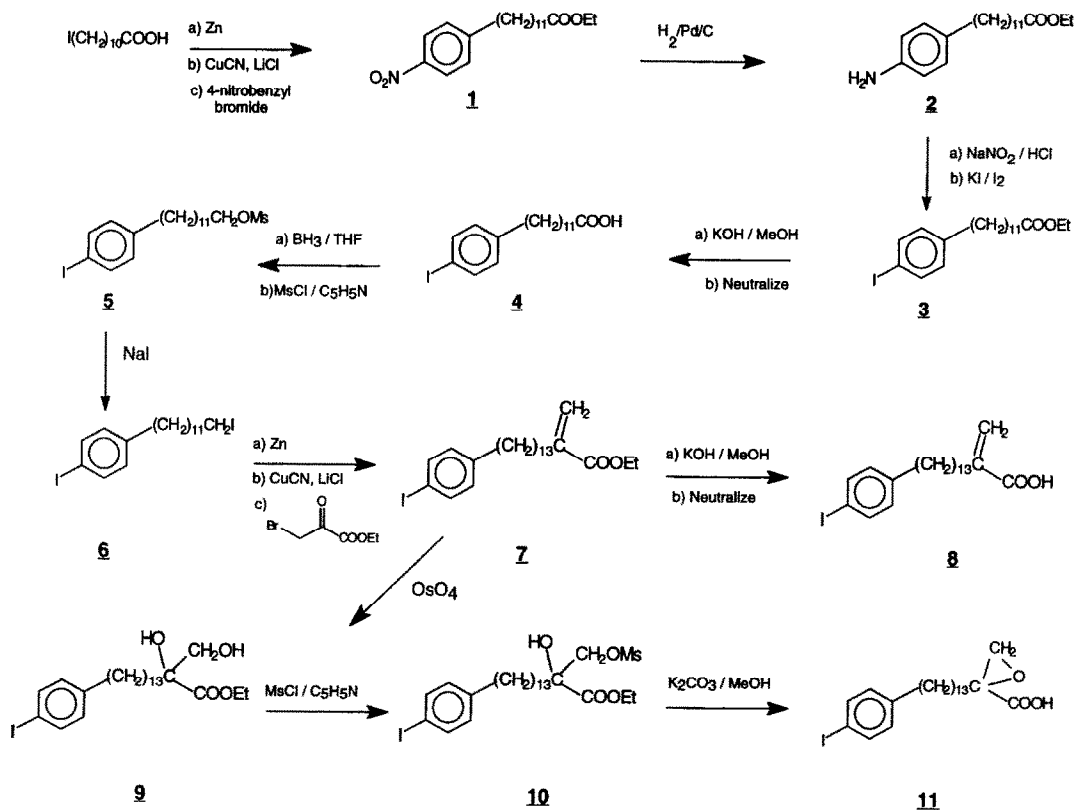


Fig. 1. Mitochondrial fatty acid transport and oxidation.



Scheme 1.

the desired glycidic acid, was also of interest since it, too, would have the capacity to bind irreversibly to enzymes involved in fatty acid metabolism. This report describes the synthesis (Scheme 1) and preliminary biodistribution and imaging analyses of two such fatty acid analogs.

Materials and Methods

All starting materials for chemical syntheses were purchased from Aldrich Chemical Co., Milwaukee, WI. Prior to use, tetrahydrofuran (THF) and benzene were dried over sodium metal with benzophenone as an indicator. All solvents were distilled under an argon atmosphere prior to use. 1H NMR spectra of intermediate and target compounds were obtained using a Bruker WM360 (360 MHz) spectrometer. All compounds were analyzed in solutions of $CDCl_3$. Results are reported in ppm (δ) using tetramethylsilane as an internal standard. Abbreviations used for interpretation of NMR data are as follows: s, singlet; d, doublet; t, triplet; q, quartet; quint, quintuplet; dd, doublet of doublets; dt, double of triplets; m, multiplet. IR spectra were recorded on either a Perkin-Elmer 281 or a Nicolet 5DXB Fourier transform spectrometer. Elemental analyses (Midwest Microlab, Indianapolis, IN or The University of Michigan Department of Chemistry, Ann Arbor, MI) of all compounds were deemed acceptable when experimental results were within 0.4% of the calculated

values. Melting points were determined with a Thomas-Hoover apparatus and are uncorrected. Thin layer chromatography was carried out on Analtech (Ann Arbor, MI) silica gel GHLF glass-backed plates. Visualization of compounds was by u.v. and by charring with an aqueous solution of ceric ammonium sulfate (4.7% w/v) and conc H_2SO_4 (5.6% w/v). Column chromatography was performed on either Davisil 62 silica gel (Davison Chemical, Baltimore, MD) or on silica gel media grade 60 (Fisher Scientific, Itasca, IL). High performance liquid chromatography was performed using an Altex 110A pump and a Gilson 111 u.v. detector (Middleton, WI) at 254 nm with a Vydac (Hesperia, CA) phenyl HPLC column (10 $\times$ 250 mm).

Chemical syntheses

The structures and synthetic approach of all compounds described in this section are shown in Scheme 1.

Ethyl 12-(4-nitrophenyl)dodecanoate (1). Zn (4.23 g, 64.7 mmol) in THF (3 ml) was activated by addition of 1,2-dibromoethane (0.21 mL, 2.5 mmol) and heating at 65°C for 1 min. This procedure was followed by the addition of chlorotrimethylsilane (0.26 mL, 2.0 mmol) at room temperature. The suspension was stirred at room temperature for an additional 15 min before a solution of ethyl 11-iodoundecanoate (Bergbreiter and Whitesides, 1975) (21.18 g, 62.3 mmol)

in THF (20 mL) was added. The reaction mixture was stirred at 43–45°C for 20 h, whereupon the clear solution of organozinc reagent was transferred to a round bottom flask containing CuCN (4.9 g, 54.8 mmol) and LiCl (4.65 g, 109.6 mmol, dried at 150°C) in THF (55 mL) at –10°C. The organometallic reagent was stirred for 10 min at 0°C before it was cooled to –70°C and a solution of 4-nitrobenzyl bromide (10.86 g, 45.6 mmol) in THF (25 mL) was added. The reaction mixture was allowed to gradually warm to 0°C and was kept at this temperature for an additional 3 h before a mixture of saturated ammonium chloride (100 mL) and hexane (150 mL) was added. The resulting precipitate was removed by filtration through a bed of celite and the organic layer was separated. The aqueous layer was extracted with Et₂O (2 × 100 mL) and combined with the organic layer. This solution was washed with brine and dried over anhydrous MgSO₄. The solvents were removed under reduced pressure and the resulting oily residue was purified by chromatography (300 g silica gel, 2% EtOAc in hexane) to afford 15.06 g of **1** (94.5%) which solidified upon standing (m.p. 35–37°C). IR (neat, NaCl), 3080, 2980, 2920, 2850, 1735, 1600, 1340 cm^{–1}. ¹H NMR (360 MHz, CDCl₃) δ 8.13 (d, 2H, arom.), 7.32 (d, 2H, arom.), 4.12 (q, 2H, –OCH₂–), 2.70 (t, 2H, benzylic), 2.28 (t, 2H, –CH₂COO–), 1.62 (m, 4H, –CH₂CH₂COO– and PhCH₂CH₂–), 1.25 (m, 17H, –(CH₂)₇– and –CH₃) ppm. Anal. (C₂₀H₃₁NO₄) C, H, N: 68.74, 8.94, 4.01. Found: 69.05, 9.20, 4.03.

Ethyl 12-(4-aminophenyl) dodecanoate (2). The nitro compound **1** (19.13 g, 54.7 mmol) dissolved in EtOAc (40 mL) was hydrogenated over 5% Pd/C at room temperature at an initial pressure of 50 psi for 6 h. The reaction mixture was filtered through a bed of celite before EtOAc was removed under reduced pressure to yield 17.0 g of **2** (97%). IR (neat, NaCl), 3470, 3380, 3010, 2930, 2860, 1735, 1625, 1520 cm^{–1}. ¹H NMR (360 MHz, CDCl₃) δ 6.96 (d, 2H, arom.), 6.62 (d, 2H, arom.), 4.12 (q, 2H, –OCH₂–), 3.18 (s, 2H, amine), 2.48 (t, 2H, benzylic), 2.28 (t, 2H, –CH₂COO–), 1.56 (m, 4H, –CH₂CH₂COO– and PhCH₂CH₂–), 1.25 (m, 17H, –(CH₂)₇– and –CH₃) ppm. Anal. (C₂₀H₃₃NO₂) C, H, N: 75.19, 10.41, 4.38. Found: 74.95, 10.54, 4.26.

Ethyl 12-(4-iodophenyl) dodecanoate (3). The amine **2** (17.0 g, 53.2 mmol) was dissolved in glacial acetic acid (100 mL) and treated with conc HCl (11 mL) which led to formation of a precipitate. This suspension was cooled to –10°C and a solution of NaNO₂ (4.04 g, 58.5 mmol) in water (18 mL) was added in several portions over 5 min. The reaction mixture was stirred at 0°C for 50 min. A solution of KI/I₂/H₂O (11.0 g, 66.5 mmol/8.1 g, 31.92 mmol/33 mL) was added dropwise over 1 h and stirring was continued overnight at room temperature. To the reaction mixture was added Et₂O (250 mL) and water (100 mL). The Et₂O layer was separated and washed sequentially with a 10% aqueous solution

of Na₂S₂O₄, a saturated solution of NaHCO₃ (until basic) and brine. The solution was subsequently dried over anhydrous MgSO₄. The solvent was removed under reduced pressure and the resulting oily residue was purified by chromatography (silica gel, 2% EtOAc in hexane) to yield 17.77 g of **3** (77.2%) as a clear oil. IR (neat, NaCl), 3020, 2980, 2920, 2825, 1735, 1480, 1460 cm^{–1}. ¹H NMR (360 MHz, CDCl₃) δ 7.57 (d, 2H, arom.), 6.92 (d, 2H, arom.), 4.12 (q, 2H, –OCH₂–), 2.53 (t, 2H, benzylic), 2.28 (t, 2H, –CH₂COO–), 1.59 (m, 4H, –CH₂CH₂COO– and PhCH₂CH₂–), 1.25 (m, 17 H, –(CH₂)₇– and –CH₃) ppm. Anal. (C₁₉H₃₁IO₂) C, H: 54.54, 7.47. Found: 54.62, 7.72.

12-(4-iodophenyl) dodecanoic acid (4). A solution of ester **3** (17.5 g, 40.66 mmol) and KOH (5.6 g, 100 mmol) in methanol (200 mL) was heated at 70°C for 5 h. Methanol was removed under reduced pressure and the resulting solid was dissolved in water. Acidification to pH 2 by addition of 1 N HCl caused the product to precipitate. The white solid was collected by filtration and recrystallized from ethanol and water to yield 16.0 g of **4** (97.8%) as white needles (m.p. 84–86°C). IR (KBr), 3100, 3020, 2900, 2920, 1690, 1460, 1400 cm^{–1}. ¹H NMR (360 MHz, CDCl₃) δ 7.58 (d, 2H, arom.), 6.92 (d, 2H, arom.), 2.53 (t, 2H, benzylic), 2.35 (t, 2H, –CH₂COO–), 1.60 (m, 4H, –CH₂CH₂COO– and PhCH₂CH₂–), 1.25 (m, 17H, –(CH₂)₇–) ppm. Anal. (C₁₈H₂₇IO₂) C, H: 53.74, 6.76. Found: 54.13, 6.91.

12-(4-iodophenyl) dodecyl methanesulfonate (5). To the acid **4** (15.5 g, 38.5 mmol) in Et₂O (100 mL) at –10°C was added BH₃·THF (46.2 mL, 46.2 mmol) slowly over 30 min. The reaction mixture was allowed to gradually warm to room temperature and stir for an additional 2 h. The reaction was quenched by addition of water at 0°C. The organic layer was separated and the aqueous layer was extracted with Et₂O (2 × 50 mL). The combined organic solution was washed with brine and dried over anhydrous MgSO₄. The solvents were removed under reduced pressure to afford 13.8 g of crude alcohol. Without further purification, this product (13.8 g, 35.4 mmol) was dissolved in anhydrous pyridine (40 mL), cooled to 0°C and methane sulfonylchloride (3.9 mL, 49.5 mmol) was added dropwise over 10 min. The reaction mixture was stirred at 0°C for 4 h and then quenched by addition of 6 N HCl (pH 2). The reaction mixture was extracted with Et₂O (3 × 50 mL) and the Et₂O solution was washed with a saturated solution of NaHCO₃, brine and then dried over anhydrous MgSO₄. Removal of Et₂O under reduced pressure and recrystallization of the resulting solid from EtOH/H₂O furnished 10.7 g (65% overall) of the mesylate **5** (m.p. 53–54°C). ¹H NMR (360 MHz, CDCl₃) δ 7.58 (d, 2H, arom.), 6.92 (d, 2H, arom.), 4.22 (q, 2H, –CH₂OMs), 3.00 (s, 3H, –CH₃), 2.53 (t, 2H, benzylic), 1.75 (quin, 2H, –CH₂CH₂OMs), 1.56 (quin, 2H, PhCH₂CH₂–), 1.26 (m, 16H, –(CH₂)₈–) ppm.

Anal. ($C_{19}H_{31}IO_3S$) C, H, I: 48.93, 6.70, 27.21. Found: 48.73, 6.77, 27.12.

12-(4-iodophenyl) dodecyl iodide (6). A solution of mesylate **5** (4.4 g, 8.66 mmol) and sodium iodide (1.6 g, 10.4 mmol) in acetone (40 mL) was heated at 50°C for 4 h. The reaction mixture was cooled at room temperature, acetone was removed *in vacuo* and water was added to the residue. The aqueous layer was extracted with Et_2O (3 $\times$ 50 mL) and the Et_2O solution was washed with brine and dried over anhydrous $MgSO_4$. The solvent was removed under reduced pressure and the crude product was purified by chromatography (100 g silica gel, hexane) to afford 3.44 g (80%) of pure **6** (mp 37.39°C). 1H NMR (360 MHz, $CDCl_3$) δ 7.58 (d, 2H, arom.), 6.92 (d, 2H, arom.), 3.18 (t, 2H, $—CH_2I$), 2.53 (t, 2H, benzylic), 1.79 (quin, 2H, $—CH_2CH_2I$), 1.54 (quin, 2H, $PhCH_2CH_2—$), 1.26 (m, 16H, $—(CH_2)_8—$) ppm. Anal. ($C_{18}H_{28}I_2$) C, H: 43.39, 5.66. Found: 43.52, 5.76.

Ethyl 15-(4-iodophenyl)-2-methylene pentadecanoate (7). To a suspension of zinc powder (0.46 g, 7.03 mmol) in THF (1 mL) at 65°C was added 1,2-dibromoethane (50 mg, 0.26 mmol). After 1 min the reaction mixture was cooled to room temperature. The suspension was treated with chlorotrimethylsilane (24 mg, 0.22 mmol) and stirred for 15 min. A solution of iodide **6** (3.37 g, 6.67 mmol) in THF (3 mL) was added to the reaction and stirring was continued for 20 h at 45°C. The organozinc reagent was transferred to a round bottom flask containing $CuCN$ (0.53 g, 5.95 mmol) and $LiCl$ (0.5 g, 11.9 mmol) in THF (6 mL) at $-10^\circ C$. The organometallic reagent was stirred for 10 min at 0°C before it was cooled to $-20^\circ C$ and treated with ethyl 2-bromomethyl acrylate (1.0 g, 5.17 mmol). The reaction mixture was stirred for 3 h at $-20^\circ C$ and 2 h at 0°C before quenching by addition of a saturated solution of ammonium chloride (50 mL). The aqueous mixture was extracted with $EtOAc$ (3 $\times$ 50 mL). The $EtOAc$ solution was washed with brine and dried over anhydrous $MgSO_4$. $EtOAc$ was removed under reduced pressure and the resulting oily residue was purified by chromatography (60 g silica gel, 2% $EtOAc$ in hexane) to yield 1.7 g (68%) of **7** as a clear oil. IR (neat, $NaCl$) 3020, 2920, 2850, 1715, 1650 cm^{-1} . 1H NMR (360 MHz, $CDCl_3$) δ 7.58 (d, 2H, arom.), 6.92 (d, 2H, arom.), 6.11 (s, 1H, vinyl), 5.50 (s, 1H, vinyl), 4.12 (q, 2H, $—OCH_2—$), 2.53 (t, 2H, benzylic), 2.29 (t, 2H, allylic), 1.56 (quin, 2H, $PhCH_2CH_2—$), 1.44 (quin, 2H, homoallylic), 1.28 (m, 21H, $—(CH_2)_9—$ and $—CH_3$) ppm. Anal. ($C_{24}H_{37}IO_2$) C, H: 59.50, 7.70. Found: 59.66, 7.79.

15-(4-iodophenyl)-2-methylene pentadecanoic acid (8). A solution of ester **7** (200 mg, 0.41 mmol) and KOH (60 mg, 1.0 mmol) in $MeOH$ (10 mL) was refluxed for 4 h. $MeOH$ was removed under reduced pressure and the resulting solid was dissolved in water and acidified with 1 N HCl (pH 2). The precipitated product was removed by filtration. The crude product was purified by chromatography (8 g silica gel, 4% $EtOAc$ in hexane) to afford 78 mg (41%) of **8**.

An analytically pure sample was obtained by recrystallization of $EtOH/H_2O$ (m.p. 65–68°C). 1H NMR (360 MHz, $CDCl_3$) δ 7.58 (d, 2H, arom.), 6.92 (d, 2H, arom.), 6.28 (s, 1H, vinyl), 5.64 (s, 1H, vinyl), 2.53 (t, 2H, benzylic), 2.29 (t, 2H, allylic), 1.57 (quin, 2H, $PhCH_2CH_2—$), 1.46 (quin, 2H, homoallylic), 1.25 (m, 18H, $—(CH_2)_8—$) ppm. Anal. ($C_{22}H_{33}IO_2$) C, H, I: 57.89, 7.29, 27.81. Found: 57.82, 7.42, 27.81.

^{125}I -15-(4-iodophenyl)-2-methylene pentadecanoic acid (^{125}I -8). To a vial was added **8** (40 μg , 0.09 μmol), ascorbic acid (4 mg), $CuSO_4$ (2 μg), DMPU (20 μL) and ^{125}I - NaI (3.0 μL , 1.5 mCi). The reaction mixture was heated at 160°C for 1 h. Purification was accomplished by chromatography (silica gel, benzene: $MeOH$:acetic acid 95:5:0.05) and the purity was confirmed by TLC (silica gel plates, benzene: $MeOH$:acetic acid 95:5:0.05, R_f = 0.23) and HPLC (Vydac phenyl column, $MeOH:H_2O$:acetic acid 70:30:0.05, t_r = 26 min). The solvent was evaporated under a stream of nitrogen to afford 1.3 mCi (87%) of ^{125}I -**8**.

Ethyl 2-hydroxy-2-hydroxymethyl-15-(4-iodophenyl)-2-methylene pentadecanoate (9). To a solution of *N*-methylmorpholine *N*-oxide (150 mg, 1.22 mmol) in water (3 mL) was added acrylate **7** (500 mg, 1.03 mmol) in acetone (2 mL) followed by OsO_4 (4% aqueous solution, 0.6 mL, 0.1 mmol) and the reaction mixture was stirred at room temperature for 30 h. A slurry of $Na_2S_2O_4$ (2.5 g) and florisil (4 g) in water (10 mL) was added to the reaction mixture and stirring was continued for another 2 h. The reaction mixture was filtered through celite. The filtrate was acidified with 1 N HCl (pH 2) and extracted with $EtOAc$ (3 $\times$ 10 mL). The $EtOAc$ solution was washed with saturated $NaHCO_3$ and brine and then dried over anhydrous $MgSO_4$. $EtOAc$ was removed and the resulting residue was recrystallized from hexane to afford 450 mg (84%) of **9** (m.p. 75.5–76.5°C). IR (KBr) 3500, 3400, 2910, 2840, 1720, 1480, 1465 cm^{-1} . 1H NMR (360 MHz, $CDCl_3$) δ 7.58 (d, 2H, arom.), 6.92 (d, 2H, arom.), 4.28 (m, 2H, $—OCH_2—$ ester), 3.78 (d, 1H, J = 11.2 Hz, $—C(OH)CH_2OH$), 3.58 (d, 1H, J = 11.2 Hz, $—C(OH)CH_2OH$), 2.53 (t, 2H, benzylic), 1.60 (m, 6H, $PhCH_2CH_2—$ and $—CH_2(OH)CH_2OH—$), 1.25 (m, 23 H, $—(CH_2)_{10—}$ and CH_3) ppm. Anal. ($C_{24}H_{39}IO_4$) C, H: 55.67, 7.58. Found: 55.67, 7.68.

Ethyl 2-hydroxy-2-(*O*-methylsulfonate)-hydroxymethyl-15-(4-iodophenyl) pentadecanoate (10). To a solution of diol **9** (420 mg, 0.81 mmol) in anhydrous pyridine (5 mL) at 0°C was added methanesulfonyl chloride (110 mg, 0.97 mmol). The reaction mixture was stirred at 0°C for 4 h and then quenched by addition of 6 N HCl (pH 2). The reaction mixture was extracted with $EtOAc$ (3 $\times$ 20 mL). The $EtOAc$ solution was washed with saturated solution of $NaHCO_3$ and brine and then dried over anhydrous $MgSO_4$. $EtOAc$ was removed *in vacuo* and the resulting solid residue was recrystallized from $EtOH/H_2O$ to provide 400 mg (82%) of **10** (m.p. 73–75°C). 1H

NMR (360 MHz, CDCl_3) δ 7.58 (d, 2H, arom.), 6.93 (d, 2H, arom.) 4.39 (d, 1H, $J = 10.4$ Hz, $-\text{CH}_2\text{OMs}$), 4.30 (q, 2H, $-\text{OCH}_2\text{CH}_2-$), 4.24 (d, 1H, $J = 10.4$ Hz, $-\text{CH}_2\text{OMs}$), 3.04 (s, 3H, CH_3SO_3-), 2.53 (t, 2H, benzylic), 1.65 (m, 4H, $\text{PhCH}_2\text{CH}_2-$ and $-\text{CH}_2\text{C}(\text{OH})-$), 1.27 (t, 3H, OCH_2CH_3), 1.24 (m, 18H, $-(\text{CH}_2)_9-$) ppm. Anal. ($\text{C}_{25}\text{H}_{40}\text{IO}_6$) C, H: 50.42, 6.77. Found: 50.80, 7.02.

15-(4-iodophenyl)tridecylglycidic acid (II). The mesylate **10** (390 mg, 0.65 mmol) and K_2CO_3 (200 mg, 1.45 mmol) in aqueous methanol were refluxed for 1 h. Methanol was removed *in vacuo* and the resulting solid residue was dissolved in saturated citric acid solution (20 mL) and the product was washed with water and brine and then dried over anhydrous MgSO_4 . Removal of EtOAc and recrystallization of remaining solid from EtOH/ H_2O afforded glycidic acid **11** (165 mg, 53.4% m.p. 95–96°C). IR (KBr) 3450, 3010, 2920, 2850, 1710, 1690, 1490, 1460 cm^{-1} . ^1H NMR (360 MHz, CDCl_3) δ 7.58 (d, 2H, arom.), 6.92 (d, 2H, arom.), 3.07 (d, 1H, $J = 5.6$ Hz, epoxide), 2.87 (d, 1H, $J = 5.6$ Hz, epoxide), 2.53 (t, 2H, benzylic), 2.14 (m, 1H, $-\text{CH}_2\text{C}(\text{CH}_2\text{O})-$), 1.56 (m, 2H, $\text{PhCH}_2\text{CH}_2-$), 1.27 (m, 20H, $-(\text{CH}_2)_{10}-$) ppm. Anal. ($\text{C}_{22}\text{H}_{33}\text{IO}_3$) C, H, I: 55.93, 7.04, 26.87. Found: 55.73, 7.14, 26.95.

^{125}I -15-(4-iodophenyl)tridecylglycidic acid (^{125}I -II). A V-shaped vial containing the ethyl ester of **11** (100 μg , 0.21 μmol), CuBr (10 μg , 15 μL in DMPU) and ^{125}I -NaI (2.0 μL , 1.0 mCi in anhydrous acetonitrile) was immersed for 45 min in an oil bath preheated to 160°C. The ethyl ester of **11** was prepared by treating **10** in dry tetrahydrofuran with 2 equivalents of sodium hydride at room temperature. The reaction mixture was transferred to a column (3 g silica gel) and was eluted with hexane:EtOAc 95:5. The fractions containing most of the radioactivity were analyzed for purity by radio TLC (silica gel plates, hexane:EtOAc 95:5, $R_f = 0.20$). The pure fractions were combined and reduced to dryness under a gentle stream of nitrogen. The ester was transferred to a second V-shaped flask and a solution of KOH in 95% EtOH (50 μg /40 μL) was added. The reaction was stirred at room temperature for 5 h. The EtOH was removed under a gentle stream of nitrogen and an aqueous citric acid solution (90 μg /30 μL) was added. The aqueous solution was extracted with EtOAc (5 $\times$ 20 μL) and the extract was directly transferred to a column (3 g silica gel) and was eluted with benzene:MeOH 95:5. The fractions containing most of the radioactivity were analyzed for purity by radio TLC (silica gel plates, benzene:MeOH:acetic acid 95:5:0.5, $R_f = 0.19$) and HPLC (Vydac phenyl column, MeOH: H_2O :acetic acid 73:27:0.075, $t_r = 11.5$ min). The pure fractions were combined and reduced to dryness under a gentle stream of nitrogen to afford 70 μCi (6% overall) of ^{125}I -II.

^{125}I -(4-iodophenyl)pentadecanoic acid (^{125}I -IPPA). A V-shaped vial containing IPPA (50 μg , 0.11 μmol), ascorbic acid (4 mg), CuSO_4 (2 μg), DMPU (20 μL)

and ^{125}I -NaI (2.0 μL , 1.0 mCi) was immersed for 45 min in an oil bath preheated to 160°C. The reaction mixture was transferred to a column (3 g silica gel) and was eluted with benzene:MeOH:acetic acid 95:5:0.5. The fractions containing most of the radioactivity were analyzed for purity by radio TLC (silica gel plates, benzene:MeOH:acetic acid 95:5:0.5, $R_f = 0.21$) and HPLC (Vydac phenyl column, MeOH:water:acetic acid 70:30:0.05, $t_r = 19$ min). The pure fractions were combined and reduced to dryness under a gentle stream of nitrogen to furnish 840 μCi (84%) of ^{125}I -IPPA.

Formulation of compounds

Compounds administered for tissue distribution studies were transferred to a vial and solvent was removed by evaporation under a stream of nitrogen. Absolute ethanol (10% of total final volume) was added to the compound and mixed well by vortex action. A solution of 2% bovine serum albumin in water was added and gently mixed by swirling. If necessary, the solution was heated to 37°C to dissolve compound completely. The solution was filtered through a 0.22 μm filter prior to use. Assays of the specific activity of the formulated dose were made for each formulation using a Searle γ -counter. All compounds were prepared and used on the same day.

Tissue distribution studies

Female Sprague–Dawley rats, 200–250 g (Charles River, Portage, MI) were used in all studies. Rats were fasted with free access to water for the 12 h prior to injection of radiolabeled compounds. Rats were anesthetized with a mixture of ketamine and rompun (87 mg/kg ketamine and 13 mg/kg rompun) prior to administration of the radiolabeled compounds. Approximately 5 μCi total radioactivity (0.1–0.3 mL of compound) were administered by tail vein injection. At 1, 2, 5, 15, 30, 60 or 120 min following injection, rats were sacrificed by exsanguination via the abdominal aorta. The heart, liver and lungs were quickly removed, rinsed of excess blood in saline and blotted dry. Whole organ weights were taken for liver and heart. The blood was transferred to a Vacutainer brand blood collection tube (Becton Dickinson, Rutherford, NJ) containing heparin. Duplicate blood samples were assayed for total radioactivity and the remaining blood was centrifuged at 800 g for 10 min to collect plasma. Samples of both plasma and tissue homogenates from each organ were weighed into tared cellulose acetate capsules and assayed immediately for total radioactivity using a Searle 1185 γ -counter (84–86% efficiency for ^{125}I). Results are expressed as the mean % administered dose/g tissue for 3 animals at each time point.

Scintigraphic imaging in rabbits

Scintigraphic imaging of female rabbits was performed using a Siemens LEM Mobile Gamma

Camera (Siemens Corporation, Hoffman Estates, IL) with a high-sensitivity-low energy collimator. Fasted, anesthetized rabbits (40 mg/kg ketamine, 10 mg/kg rompun) were placed in a supine position directly on the inverted camera face and a 250 μ Ci bolus dose of 125 I-IPPA or 125 I-**8** was administered by ear vein injection. Images of the heart and abdomen were accumulated at 5 min intervals for up to 30 min after injection of compound.

Results and Discussion

Chemistry

Our synthetic approach to the desired radioiodinated fatty acid analogs capitalized upon a recently discovered cross-coupling reaction between aliphatic halides (Yeh and Knochel, 1989; Knochel *et al.*, 1989). The key reactant is prepared by reaction of an alkyl iodide with zinc powder followed by addition of copper cyanide to form an organometallic complex which is capable of attacking suitably substituted benzyl halides.

As outlined in Scheme 1, the organozinc-copper reagent made from ethyl 11-iodoundecanoate (Bergbreiter and Whitesides, 1975) was cross-coupled with 4-iodophenyl-nitrobenzyl bromide to afford ethyl 12-(4-nitrophenyl) dodecanoate **1** in 94.5% yield. Catalytic hydrogenation afforded the amino-compound **2**, which was then converted to ethyl 12-(4-iodophenyl) dodecanoate **3** in 77% overall yield. Base hydrolysis of this ester followed by diborane reduction of the resulting acid **4** produced the intermediate alcohol which was converted directly to the mesylate **5**. 12-(4-iodophenyl)-1-iododecane **6** was made from **5** and cross-coupled with ethyl 2-bromomethylacrylate to afford ethyl 15-(4-iodophenyl)-2-methylene pentadecanoate **7** in 68% yield. Base hydrolysis furnished the desired 15-(4-iodophenyl)-2-methylene pentadecanoic acid **8**. Alternatively, dihydroxylation of **7** by treatment with OsO₄ furnished the glycol **9** which upon mesylation **10** and treatment with base produced the desired glycidic acid **11**. Various attempts to obtain **11** directly via epoxidation of **8** (or ester **7**) were

Table 1. Tissue distribution (expressed as % dose/g tissue) following i.v. administration of 125 I IPPA in fasted female Sprague-Dawley rats ($n = 3$ for each time point)

Tissue	Time after administration				
	1 min	2 min	5 min	15 min	2 h
Blood	1.387	1.066	1.526	1.308	1.350
SEM	0.112	0.146	0.130	0.088	0.141
Inj. Site	0.154	0.286	3.327	0.816	1.554
SEM	0.032	0.015	2.946	0.313	1.205
Liver	7.869	9.393	7.382	6.144	2.434
SEM	0.303	1.223	0.555	0.391	0.262
Lung	2.230	2.156	1.622	1.301	1.292
SEM	0.243	0.136	0.065	0.052	0.120
Plasma	1.841	1.690	2.316	2.331	2.449
SEM	0.114	0.221	0.095	0.171	0.124
Heart	5.019	5.537	3.001	1.652	1.071
SEM	0.093	0.778	0.181	0.109	0.101
Thyroid	3.016	2.320	5.830	2.196	23.863
SEM	0.474	0.198	0.861	0.363	0.959
Heart/Blood	3.62	5.20	1.97	1.26	0.79
Liver/Blood	5.67	8.81	4.84	4.70	1.80

unsuccessful presumably due to the susceptibility of the aromatic iodine to oxidation.

Radioiodination of the final products was accomplished by isotopic exchange with sodium 125 I. The method was a modification of the procedure of Mertens (Mertens *et al.*, 1984) and involved exchange using CuSO₄ in the presence of ascorbic acid as the catalyst. We found the most appropriate solvent for our purposes to be 1,3-dimethyl-3,4,5,6-tetrahydro-2-(1H)-pyrimidinone (DMPU). Radiochemical yields in excess of 80% were obtained for the radioiodination of IPPA and the α -methylene acid **8**. However, the glycidic acid analog **11** was unstable to such conditions and could only be obtained in low yield by radioiodination of the ester followed by hydrolysis to the acid.

Biodistribution in rats

IPPA was first described by Machulla (Machulla *et al.*, 1978) as a myocardial imaging agent and has been used clinically for many years. Consequently, IPPA was chosen as the standard compound against which efficiency and biodistribution profiles of the new compounds would be compared.

Table 2. Tissue distribution (expressed as % dose/g tissue) following i.v. administration of compound 125 I **8** in fasted female Sprague-Dawley rats ($n = 3$, for 2 min $n = 2$)

Tissue	Time after administration						
	1 min	2 min	5 min	15 min	30 min	1 h	2 h
Blood	1.772	1.473	0.902	1.277	1.662	1.161	1.109
SEM	0.294	0.186	0.175	0.200	0.258	0.191	0.041
Inj. Site	0.258	0.162	0.282	0.376	0.347	1.282	0.320
SEM	0.062	0.033	0.089	0.098	0.073	1.018	0.044
Liver	8.479	7.740	7.996	7.791	5.450	3.694	2.451
SEM	0.465	0.072	0.264	0.520	0.237	0.066	0.139
Lung	1.956	1.239	1.213	1.363	1.170	0.947	0.800
SEM	0.220	0.091	0.053	0.265	0.097	0.032	0.009
Plasma	2.462	2.023	1.465	2.297	2.878	1.999	1.991
SEM	0.404	0.305	0.289	0.391	0.448	0.136	0.064
Heart	4.746	5.022	3.627	2.135	2.204	1.338	1.173
SEM	0.384	1.158	0.805	0.369	0.114	0.020	0.053
Thyroid	1.083	0.913	1.305	2.022	2.224	3.147	3.894
SEM	0.138	0.276	0.361	0.142	0.487	0.313	1.209
Heart/Blood	2.68	3.40	4.00	1.67	1.33	1.12	1.06
Liver/Blood	4.78	5.25	8.86	6.10	3.28	3.18	2.21

Compound **8** is very similar in structure to IPPA, with the only difference being the presence of a methylene group on the α -carbon. Incorporation of the methylene group was intended to increase myocardial retention either by directly inhibiting β -oxidation of the acid or by facilitating irreversible binding of the acid to intracellular macromolecules through a Michael-type addition. As indicated from tissue distribution data (Tables 1–2), the presence of the methylene group did not significantly extend the retention time of the compound in the heart or liver relative to IPPA. However, the heart/blood ratio of the methylene derivative of IPPA was twice that of IPPA alone at 5 min post injection. In fact, the heart/blood ratios of the methylene analog remained substantially higher than those of IPPA alone at all timepoints tested after 5 min.

Another interesting feature of the α -methylene analog of IPPA is its apparent ability to resist deiodination. At 2 h following injection of IPPA, the thyroid contained almost 24% of the injected dose/g tissue (Table 1). By comparison, the α -methylene derivative resulted in a thyroid radioiodine concentration of just under 4% of the injected dose/g tissue

at the same timepoint (Table 2). Thus, the iodine on the α -methylene analog of IPPA seems much more stable to deiodination processes and therefore may offer a benefit over IPPA.

Unlike the α -methylene analog, the glycidic acid analog of IPPA **11** was accumulated relatively poorly by the heart. For example, at 1 and 2 min following administration, the concentrations of compound in the rat heart were only 1.8 and 1.7% dose/g, respectively. Extraction by the liver, on the other hand, was very high, reaching levels of 9.5% dose/g at 15 min post-injection (data not shown). As a result of the poor myocardial uptake of compound **11**, no imaging studies were undertaken with this agent.

Scintigraphic imaging in rabbits

Because the biodistribution profile for the α -methylene derivative of IPPA provided some improvements over IPPA alone, follow-up *in vivo* scintigraphic studies were performed in rabbits. For these studies, 250 μ Ci of 125 I-IPPA and 125 I-**8** were administered separately to female rabbits via the ear vein. Immediately following administration, the thoracic cavity was imaged for six consecutive 5 min

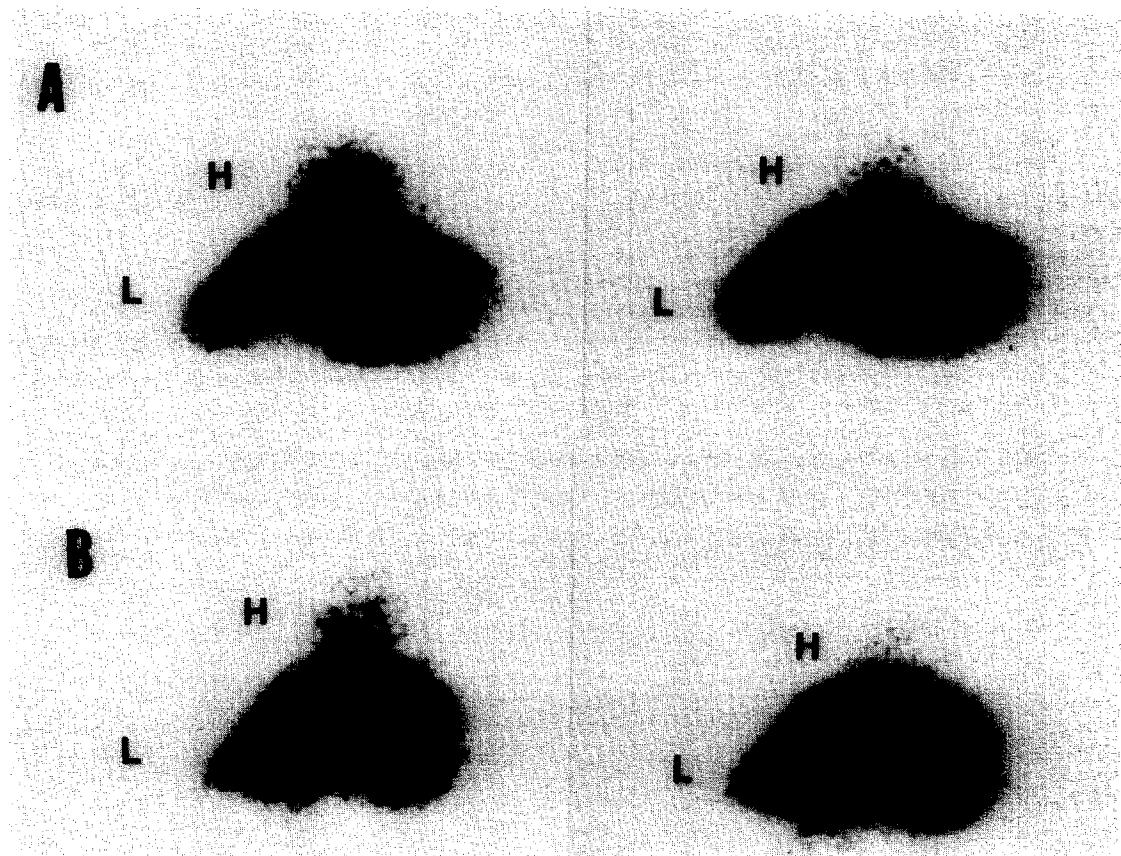


Fig. 2. (A) Upper body scintigraphic images of a female rabbit after i.v. administration of 250 μ Ci of 125 I-IPPA. Time frames are 5 min (left) and 10 min (right). H = heart and L = liver. Total counts were 148,708 and 153,336 at 5 and 10 min respectively. (B) Upper body scintigraphic images of a female rabbit after i.v. administration of 250 μ Ci of 125 I-**8**. Time frames are 5 min (left) and 10 min (right). H = heart and L = liver. Total counts were 138,475 and 139,395 at 5 and 10 min, respectively.

intervals. Scans for only the 5 and 10 min intervals are shown in Fig. 2. Although the biodistribution data would have suggested the α -methylene analog to be essentially equivalent to IPPA, this was not reflected in the scans, which showed IPPA to be somewhat superior.

Conclusions

Using organozinc cross-coupling reactions, each of the target compounds were successfully prepared. Of the two target compounds designed to serve as myocardial imaging agents, one appeared to possess qualities equal to or slightly better than those observed for IPPA. The α -methylene derivative of IPPA displayed a consistently higher heart to blood ratio at later timepoints and a substantially lower degree of thyroid accumulation than did IPPA alone. With respect to imaging efficacy, however, the α -methylene analog of IPPA was found to be no better than IPPA itself. On the basis of these results, neither the α -methylene or glycidic acid analogs of IPPA appear to warrant further investigation.

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